Sahaj Bhadja

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Summary

Full-stack software engineer with 2+ years shipping **AI products, LLM systems, real-time collaboration, and enterprise workflows** using React, TypeScript, Node.js, Python, scalable APIs, and cloud infrastructure.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Restructured fault-tolerant asynchronous LLM orchestration using FastAPI background workers, processing 1,000+ AI jobs with explicit state transitions and failure isolation while increasing throughput 3x under concurrent production workloads.
- Owned cloud data workflows from design through deployment, monitoring, and iteration, processing 5,000+ monthly jobs with reconciliation, error handling, and CI/CD releases while sustaining 99%+ reliability at scale.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Shipped a full-stack AI grading product with React and Python, serving 150+ instructors and 10,000+ daily requests while reducing feedback turnaround 60% through customer-focused improvements to instructor workflows.
- Built concurrent LLM processing across OpenAI, Anthropic, and Qwen models, adding secure file ingestion, fallbacks, and usage monitoring to consistently maintain reliable API performance during peak workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Developed a GPT-based Decision Transformer and end-to-end Python pipelines processing 100,000+ trajectory logs, improving long-horizon planning 18% through reliable, iterative model design, simulation, and performance evaluation at scale.
- Designed spatial-temporal attention across ego, road, and nearby-agent signals, improving closed-loop stability 15% while documenting experiments and collaborating with teammates on difficult autonomous decision-making problems at scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Owned cross-platform microservice features from requirements through deployment, collaborating with Microsoft and GitHub stakeholders while automating tests across three operating systems and increasing delivery pipeline efficiency 68%.
- Improved API performance and scalability for low-bandwidth customers, reducing response times 30%, payload sizes 25%, storage costs 15%, and technical debt 20% through pragmatic backend optimizations at scale.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built customer-facing React and Node.js features for eight modular CRM services, using WebSockets and REST APIs to synchronize two million records with 33% fewer conflicts across locations.
- Turned usage and performance insights into product improvements, reducing resource waste 42% and increasing lead conversions 15% while shipping through Docker, Jenkins, automated tests, and staged deployments.

Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, Shell, Bash
Full Stack: React, Next.js, Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets
AI Systems: LLMs, GPT-4o, Claude, Qwen, PyTorch, Transformers, prompt engineering, OCR
Data: PostgreSQL, MySQL, MongoDB, Redis, Azure SQL, vector search, FAISS, Pinecone
Engineering: Algorithms, data structures, software design, Git, CI/CD, Docker, AWS, Azure, monitoring

Projects

Big-Brain Live | *LLMs, Real-Time Collaboration*

- Built an LLaMA-2 document assistant over 50,000+ records with semantic search, conversational question answering, and real-time Convex collaboration, reducing document-analysis time 60% through low-latency synchronized workflows for users.

FinNexus Live | *TypeScript, Scalable Workflows*

- Developed a Next.js and TypeScript fintech platform supporting 50,000+ users and one million transactions, delivering real-time WebSocket updates, secure Plaid integration, advanced filtering, and Kubernetes-based scalability in production.

PiXMod Live | *AI-Native SaaS, Next.js*

- Shipped an AI-native image product using CLIP, SAM, and Stable Diffusion, processing 1,000+ edited assets with responsive Next.js workflows, scalable MongoDB storage, semantic search, and Stripe subscriptions.